1st file updated continue……

Mid Term Current File

ZOO403

Created BY CH Ismail

1. Morphological Responses?

- Changing in the of the body appearance
- Growing thicker fur or change in fur color in winter

Examples:

- Chameleon change color when it feels threatened.
- Arctic hare change its coat color from blue-grey to white for better camouflage during winters and turn again to blue-grey in spring.

2. Define Kinesis with example?

A change in activity or turning rate in response to a change in location"

Kinesis—Example:

• Kinesis in sow bugs in response to variation in humidity

• Move more rapidly in dry areas and become slow in humid areas

• They survive better in moist areas.

3. What is learning and Types of learning? **Repeated**

Learning Behaviors:

Behaviors those are variable with experience

Examples

• Habituation

• Imprinting

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Types:

1) Habituation
2) Imprinting
3) Classical conditioning
4) Trial and error learning
5) Latent learning
6) Insight
7) Reasoning
8) Cognition

4. Define taxis and also positive and negative taxis? Repeated

Taxis:

"An oriented movement toward (positive taxis) or away (negative taxis) from some stimulus"

Taxis—Examples:

River fishes (e.g. trout) swim or orient in an upstream direction (toward the current)

Positive taxis:

If the organism moves towards the stimulus the taxis are positive.

For example, flagellate protozoans of the genus Euglena move towards a light source. This reaction or behavior is called positive taxis.

Negative taxis:

If the organism moves away the stimulus the taxis are negative.

An example of negative taxis would be the fly moving away from an insect repellent.

5. Write Environmental Factors in Behavior Development?

• Genes of an individual provides instructions for behavior development.

• Expression of genes is influenced by many environmental factors.

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• Animal behavior develops by complex interactions between genetic and environmental factors.

• Environment of fertilized egg

• Animal's diet

• Animals social interactions

• Environmental conditions in animal are surrounding.

6. Significance of habituation?

Definition:

A loss of responsiveness to stimuli that convey little or no new information.

Significance of Habituation:

• An animal conserves energy and time by habituation to unimportant stimuli
• Energy and time are better spent on other important functions

Habituation—Examples

Many mammals and birds recognize alarm calls of members of their species.

7. Acclimatization?

Acclimatization is an adaptive change within an individual animal.
Happens due to chronic exposure to new, naturally occurring environmental conditions
.

Example:

An animal migrates to a high altitude area faces low oxygen partial pressure.

8. Insight Learning example?

Insight is immediate and clear learning. Insight occurs when animals recognize relationships that can help them solve new problems.

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Examples of insight learning include:

- Any time you'd had an epiphany in the shower or on a walk around the block
- Discovering the solution to a problem only after you've left the office and drove home
- Dreaming of the right decision to make or solution to apply.

9. Habitat of dolphin? 2

• Dolphins are marine mammals closely related to whales and porpoises.
• There are almost forty species of dolphins.
• Dolphins are among the intelligent animals, and their often friendly appearance and seemingly playful attitude have made them popular in human culture.

10. Types of Animal behavior? 2

There are 2 basic types of behavior.
1. Innate behaviors or Inherent behavior
2. Learned behaviors or acquired behavior

11. Name animals which researchers used? 5

Researchers have examined animal cognition in:

- Mammals (primates, cetaceans, elephants, dogs, cats, horses, cattle, raccoons and rodents),
- Birds (including parrots, fowl and pigeons),
- Reptiles (lizards, snakes, and turtles)
- Fishes
- Invertebrates (cephalopods)

12. Explain instinctive with nest building in birds_ an example of instinctive or innate behavior? 5

Nest building in birds_ an example of instinctive or innate behavior:
• A bird nest is the spot in which a bird lays and incubates its eggs and raises its young.
• For most birds, when a male and female form a breeding pair nest building activities will follow.

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• Nest building is an instinctive behavior as birds have innate ability to build nests.
• A nest site is first selected and nest building begins.
• This can involve both birds as a pair or either sex alone.
• Different species of birds build different nests. Some nests are very simple constructions, whilst others are more complex.

Birds build nests to:

- To lay their eggs in and keep them safe
- To incubate their eggs
- To raise their chicks
- To protect their eggs and chicks from predators and adverse weather conditions.

Nests are built in:

- Hedges, bushes and trees
- Directly on the ground
- Holes, trees or reeds by rivers and lakes
- Tunnels in riverbanks and cliffs

13. Differentiate B/W orthokinesis and klinokinesis? **Rep...**

Ortho kinesis:
It is a response that involves changes in the speed of movement of the whole body in response to stimuli like humidity, preserve and diffused light.

Klinokinesis:

The speed of locomotion remains constant but the rate at which the animal changes direction depends on the intensity of the stimulus.

14. Example of animal cognition?

Animal Cognition:

Ability of an animal's nervous system to perceive, store, process, and use information gathered by sensory receptors.

Examples of Animal Cognition:

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• Chimpanzees learn to join sticks to reach bananas
• Clark's nutcrackers can find food hidden in caches located halfway between particular landmarks.

Examples of Animal Cognition:

• Ravens showing sophisticated social cognitive skills including the formation of coalitions, considering visual perspectives of others, and directing conspecifics' attention to external referents.

15. Characteristics of innate behavior?

Define:
Innate behavior is inborn or innate sequence of activities which are predictable, species specific, genetically controlled and independent of past experience.
• Since this type of behavior follows a fixed pattern (FAP) which is predictable and found in all members of a species it is also called stereotyped behavior.

Characteristics:

• Pattern of behavior is inherited. It is passed on from parents to offspring.
• It is unlearned behavior.
• It occurs in all the members of a species hence it is species specific and predictable.
• It is not dependent on past experience as it is an inborn response to a stimulus
• It takes place in individuals even when kept in isolation away from their fellow members
• Innate behavior has high adaptability and survival value.

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16. Adaptation of green anoles? **Rep…**

- **Green Anoles use their pink dewlaps for two purposes.**

- For breeding, displaying their intent to females.
- For establishing territory among adult males. By displaying his dewlap, the lizard can tell other males that they have entered his home range.

17. Hormones and animals behavior? 5

The scientific study of the interaction between hormones and behavior is called behavioral endocrinology.
This interaction is bidirectional: hormones can influence behavior, and behavior can sometimes influence hormone concentrations.
Hormones are chemical messengers Released from endocrine glands Travel through the blood system to influence the nervous system to regulate behaviors such as:
• Aggression
• Mating
• parenting of individuals

Examples of hormones that influence behavior:

• Steroid hormones such as testosterone
• Estradiol (a common type of estrogen)
• Progesterone (a common type of progestin)
• Cortisol (a common type of glucocorticoid)

18. Fixed Action Patterns (Instincts)? Rep….

• It is a sequence of unlearned, innate behavior that is unchangeable.
• Once initiated; it is usually carried to completion.
• It is hardwired, however, not purely genetic, may improve with experience.
• FAP is triggered by an external sensory stimulus known as "sign stimulus".

Example: Communication in bees by Karl von Frisch.

Properties of Fixed Action Patterns:

• Are relatively complex motor acts
• Are all-or-none responses i.e. once initiated; continue to completion even if the stimulus is removed
• All members of a species perform a given fixed action pattern identically
• Patterns are inherited genetically
• Are exhibited perfectly without prior experience.

19. Synaptic?

The reshaping (plasticity) of nervous system occurs at synapses
• Often referred as synaptic plasticity

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• Mechanisms underlying synaptic plasticity are currently under intense research.

20. Resting membrane potential?

A “resting” neuron is not conducting a nerve impulse.
The plasma membrane of a resting neuron is polarized; the fluid on the inner side of the membrane is negatively charged with respect to the positively charged fluid outside the membrane. The difference in electrical charge between the inside and the outside of the membrane at any given point is due to the relative numbers of positive and negative ions in the fluids on either side of the membrane, and to the permeability of the plasma membrane to these ions. The difference in charge is called the resting membrane potential. All cells have such a resting potential, but neurons and muscle cells are specialized to transmit and recycle it rapidly. The resting potential is measured in millivolts (mV).

21. Animal Orientation with example? 5

Animal Orientation “Ability of an animal to determine its position in space and move predictably with respect to specific stimuli”

Animal Orientation: Examples

• Standing up after falling down (orientation against gravity)
• Orienting towards incoming light
• Acoustic orientation
• Olfactory orientation

Sensory Motor Basis of Orientation:
• Animal orientation is a complex process
• Requires integration of sensory input and coordination of motor output

22. Insight learning?

Insight is immediate and clear learning understanding that takes place without overt trial-and-error testing
• Insight occurs when animals recognize relationships that can help them solve new problems.

23. Associative learning?

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"A learning that involves making associations between experiences"
• Ability to associate one environmental feature (such as a color) with another (such as a foul taste).

Example:

• A mouse bites a brightly colored caterpillar
• It tastes bad
• Mouse avoids attacking insects of similar color in future.

Types:

• Classical conditioning
• Operant conditioning

24. Angelo's d experiments on imprinting?

• Angelo d'Arrigo extended the technique used by Konrad Lorenz.
• He noted that the flight of hang-glider is very similar to the flight patterns of migratory birds.
• Both use updrafts of thermal currents to gain altitude that then permits soaring flight over distance.

25. Write Examples of Neural Plasticity?

• Learning, memory
• Development of motor skills and habits
• Ability to respond to stimuli in diverse ways (different from fixed, programmed reflexes)
• Lies behind human intelligence

26. Innate or Inborn Behaviors describe?

• Also known as innate, inherent o instinctive behaviors.
• Complex behaviors
• Exhibited by most members of a species
• Are built-in part of nervous system at the time of birth
• Some develop at an appropriate point in maturation.
• Developmentally fixed.

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27. Spatial Learning?

"Capacity to establish a memory that reflects environment's spatial structure"

• Spatial learning enhances an organism's fitness in environment
• Locations of nest sites
• Hazards
• Food
• Prospective mates

Spatial Learning—Example

• Behavior of a female digger wasp
• Nests in small burrows dug into sand mounds.

28. Differ b/w Morphological and physiological Responses?

- **Morphological Responses:**

- changing in the of the body appearance
- Growing thicker fur or change in fur color in winter

Examples:

• Chameleon change color when it feels threatened.
• Arctic hare change its coat color from blue-grey to white for better camouflage during winters and turn again to blue-grey in spring.

- **Physiological Responses:**

- Changing the functioning of the body
- Acclimation (dilating capillaries to release heat)
- Acclimatization

29. Waggle dance? 2

• A special dance performed by a worker bee called "waggle dance"
• Workers returned to the honey comb with pollen or nectar,
• Dances constitute a language that "tells" other workers where the food is.

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30. Web making in spiders describe?

• A spider web or cobweb (from the archaic word coppe, meaning "spider") is a structure created by a spider out of proteinaceous spider silk extruded from its spinnerets, generally meant to catch its prey.

Process of web making:

• When a spider begins a web, it releases a silk thread. It anchors the thread to some object such as a branch.
• As the spider moves back and forth, it adds more threads, strengthening the web and creating a pattern.
• Lines that go from the center of the web outward are called "radial lines". They support the web.
• Threads that go around and around the web are called "orb lines."
• The main reason spiders spin webs is to catch the prey.

31. Define reflex arc?

A reflex action requires a reflex arc which consists of:

- a sensory organ (receptor),
- a sensory nerve (afferent nerve),
- the spinal cord or brain,
- an intermediate motor or efferent nerve
- a motor organ or effector

32. Advantages of Pheromones as Chemical Signal?

• Provide a simple message that can last for hours or days
• Are equally effective during night or day • May be transported over long distances by air
• Take relatively little energy to produce.

33. Pheromones and their role:

• Chemical messengers released by many animals in environment to communicate are called pheromones.
• Comprise 2 Greek words: "Pherein" (to transfer) and "Hormon" (to excite)

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Chemical Communication:

• Chemical communication through pheromones is well developed in:
• Unicellular organisms
• Insects
• Fishes
• Salamanders
• Mammals

Roles of Pheromones:

1. Recognize Members of Species
2. Reproductive Roles
3. Territorial Behaviors
4. Insect's Social Order
5. Alarm Calls
6. Repel Predators

34. What is feature detector?

The particular area of nervous system that is responsible for prey capture behavior in toads is called as feature detector.

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BEST OF LUCK

Remember Me in Your Prayers